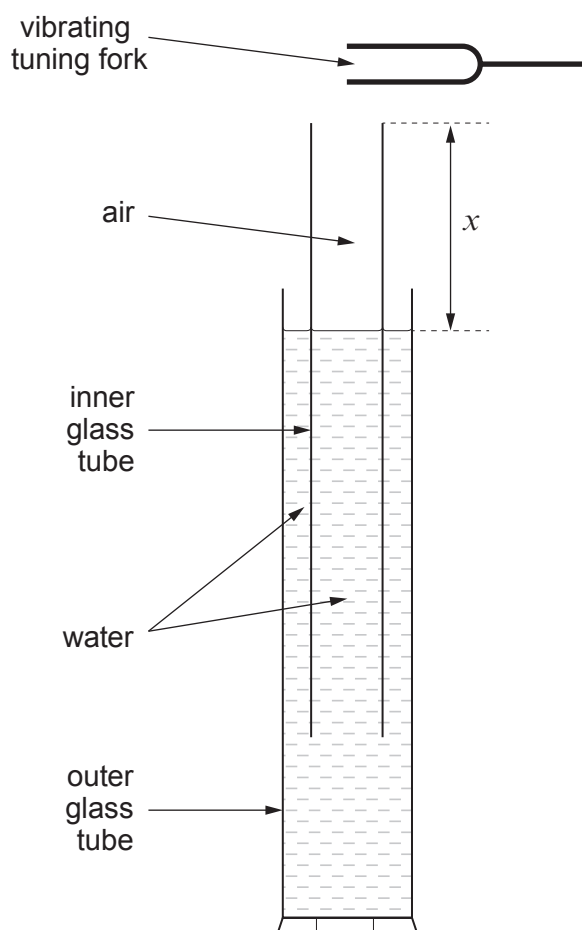


6. Matthew uses the apparatus shown to find a value for the speed of sound in air using stationary waves.



Starting with a very low value of distance x , he gradually raises the inner glass tube. At a certain value of x , the note of the tuning fork is heard loudly. Matthew then measures and records x .

He repeats the procedure several times, obtaining these results.

x / m	0.327	0.329	0.322	0.328	0.332	0.323
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The tuning fork frequency, f , is given as $256.0 \text{ Hz} \pm 1\%$.



- (a) The speed of sound in air, v_s , can be found from the equation:

$$v_s = 4fx$$

- (i) Calculate the mean value of x along with its **percentage** uncertainty. [2]

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- (ii) Hence calculate the value of v_s along with its **absolute** uncertainty. [2]

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- (b) The loud sound is due to stationary waves of frequency, f , in the air column. There is a node of displacement at the water surface and an antinode of displacement close to the open end (top) of the inner glass tube.

(i) Derive the equation:

[2]

$$v_s = 4fx$$

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- (ii) If the inner glass tube is raised considerably further a second distance is found at which a loud note of frequency, f , is heard. This again corresponds to a stationary wave in the air column. Show the positions of its nodes and antinodes of displacement **on the diagram below**.

[2]

